

The Lens is one of those rare inventions which derives its name from an edible object – lentil. That's because a double-convex lens is lentil-shaped. The oldest known lens is a 3000-year-old rock crystal called 'Nimrud Lens' or 'Layard Lens' deriving its name from the ancient city of Nimrud, now part of Iraq.



One of the most widespread uses of lenses is in aiding the correction of human eye vision. Binoculars would be second in the list while microscopes and telescopes would be the most advanced and fancy implementations. Lenses have gifted us with the ability to see as far as galaxies away and as near as the tiny cells in our bodies. We owe a lot of useful discoveries to their invention.

Memristor

Less than a decade ago, not many people were sure about the existence of memristors. Now, in mid-2013, memristors are re-defining fundamental electronics as we know them and bringing in a digital revolution of sorts. Believe us, that it's not an exaggeration.

Their story begins in 1971 when Leon Chua, an electrical engineer in his mid-30s, wrote a paper called "Memristor--the Missing Circuit Element" and predicted their discovery on mathematical grounds -- much like Mendelev predicted the missing elements of the periodic table. Chua argued that besides resistors, capacitors and inductors, memristors were the fourth basic (and until then missing) building block of an electric circuit. Forty years later, he's been proven right.

Memristors are passive components that "remember" the amount of current flowing through it. Thanks to research into memristive materials and technologies by companies like HP, Panasonic and memory manufacturers, we should be able to witness a ReRAM (Resistive RAM) module later this year. It's non-volatile, operates at lower voltages, denser than flash and quicker than DRAM. In fact, universal memory modules derived